

A PROJECT PROPOSAL ON
SajiloMart: A Ecommerce Platform



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1. Introduction

SajiloMart is a simple and innovative web-based e-commerce platform designed to make online buying, selling, order handling, and product delivery easy. This system helps buyers, sellers, warehouse staff, assistant, and delivery men work together on a single platform without needing advanced technical skills.

The platform supports five primary user roles: **Admin, Assistant, Warehouse Staff, Seller, Buyer** and **Delivery Men** each with clearly defined functionalities to support their respective workflows. This project aims to provide a smooth and fast online shopping experience for all users. It focuses on managing products, orders, inventory, and deliveries in an organized and user-friendly way. The platform is designed to give an easy shopping experience by using features like AI-based product recommendations, emergency fast delivery, AI-smart product comparison. Buyers can browse products, place orders, and track deliveries easily. Sellers and staff can manage products, stock, and orders efficiently. Using local warehouses and nearby delivery men, the system will reduce delivery time and cost. Customers will be able to get products on the same day or next day. This approach will also support local sellers and delivery workers, making the system helpful for the community.

Our system has multiple access modes, and each mode has specific roles and permissions to ensure proper and smooth e-commerce operations. They are: -

- Admin
- Assistant
- Warehouse staff
- Seller
- Buyer
- Delivery Men

2. Problem Statement

In many existing systems, people often face difficulty while shopping online. Buyers often spend too much time searching for the right product from many websites. It is difficult to find products that are available nearby or can be delivered quickly. Sellers also face problems managing their products, orders, and delivery efficiently. They cannot easily track which orders are ready, packed, or delivered. Warehouse staff struggle to manage stock and dispatch products on time. Delivery Men often do not know the exact location or status of the product, which can delay delivery.

The lack of proper coordination between buyers, sellers, warehouse staff, and delivery men creates confusion and delays. There is also no system to suggest products based on user preferences or detect feedback automatically. Because of this, customers do not get personalized shopping experience, and sellers miss important information about their products. Managing all these tasks manually is time-consuming and leads to mistakes.

Therefore, there is a need for a single platform that will connect all users, will manage products, orders, and deliveries efficiently, and will provide a better online shopping experience for everyone.

3.Objectives

3. 1 Primary Objectives

- To provide a user-friendly, organized platform for buyers, sellers, warehouse staff, and delivery men.
- To improve the online shopping experience by providing reliable and intelligent system features.

3.2 Secondary Objectives

- To manage users with different roles and permissions such as buyers, sellers, warehouse staff & delivery man.
- To design a simple and responsive interface that works on computers, tablets and mobile devices.
- To introduce an emergency fast delivery option for customers who require urgent product delivery.
- To provide AI-based product recommendations that suggest products to users based on their interests, budgets and previous activity.
- To implement a smart product comparison system that allows users to compare products based on price, features.

4. Scope and Limitations

4.1 Scope

The main intention of this project is to solve the problems faced in traditional online shopping and product delivery systems, where most processes are manual or scattered across different platforms. Many e-commerce systems exist, but they often fail to provide smooth coordination between buyers, sellers, staff, and delivery man.

In this project, a web application will be developed to manage products, orders, inventory, and deliveries in one place. Buyers can easily search and order products, track delivery status, and provide feedback. Sellers can upload products, manage prices, and monitor orders. Warehouse staff can manage stock and prepare products for delivery. Delivery men can view assigned deliveries and update delivery status in real-time.

The platform also includes AI-based features such as personalized product recommendations and review analysis to improve the shopping experience. This system is

designed to save time, reduce delivery delays, and make online shopping faster, easier, and more organized for all users.

4.2 Limitations

- The platform requires internet access to function and cannot be used offline.
- Sellers and staff must have basic knowledge of using computers or smartphones to use the system efficiently.

5. Methodology

5.1 Requirement Identification

5.1.1 Study of existing system

To understand how current online shopping platforms work, a study was conducted on two e-commerce applications: Daraz and SmartDoko. These platforms allow buyers to purchase products online from different sellers and receive them through delivery services. Studying these systems helped to understand their useful features as well as some limitations. Based on the study these were some strengths and limitations that were found.

I.Daraz

Daraz is a popular online shopping platform where customers can buy different types of products such as electronics, clothing, beauty products and household items. It connects many sellers with buyers through an online marketplace and provides delivery services to customers.

Strengths:

- Buyers can pay using different methods such as cash on delivery, debit/credit cards, and digital wallets, which makes payment convenient for users.
- It offers a wide range of products from many sellers, allowing buyers to choose from different brands, prices and product categories.

- The platform frequently provides special sales, seasonal offers and promotional discounts which attract many buyers.

Drawbacks:

- Sometimes delivery can take several days, especially when products are shipped from distant locations.
- Since many sellers use the platform, the quality of products may vary and sometimes buyers may receive products that do not fully match their expectations.

ii.SmartDoko

SmartDoko is another e-commerce platform in Nepal that provides online shopping services. It allows customers to buy different types of products and supports local sellers by giving them an online marketplace to sell their goods.

Strengths:

- SmartDoko focuses on promoting local businesses and helping them sell their products through an online platform.
- Buyers can easily browse products, place orders and receive their items through delivery services.

Drawbacks:

- Compared to larger platforms, the number of products and sellers is smaller.
- Some advanced features such as smart product comparison, intelligent recommendations or optimized delivery management are limited.

Based on the analysis of above platforms, several functional and usability gaps were identified. These platforms provide basic online shopping features but may lack advanced functionalities such as intelligent product comparison, smart product recommendations, and optimized delivery management. Therefore, our proposed system SajiloMart aims to address

these limitations by creating a more organized and efficient e-commerce platform. Our system will include multiple user roles such as Admin, Assistant, Warehouse Staff, Seller, Buyer, and Delivery Men. It will also provide features like AI-based product recommendations, smart product comparison, emergency fast delivery. This approach will help improve the overall shopping experience and make the system more efficient for both buyers and sellers.

5.1.2 Requirement Collection

i. Brainstorming:

We looked at various online shopping platforms, watched related videos such as product review videos, e-commerce app tutorials, and customer experience videos, and talked with potential users like buyers, sellers, and delivery personnel to understand their experiences, challenges, and needs. We carefully noted all the ideas and suggestions shared. After gathering this information, we selected the most useful and practical ideas to include in the project. This approach helped us consider many perspectives and choose the best features for our project.

ii. Observation:

We observed how people currently shop online and in physical stores, how sellers manage their products, and how deliveries are handled. We noted problems such as difficulty in comparing products, slow deliveries, and miscommunication between buyers and sellers. We also looked at how sellers organize their products and interact with customers. These real-life observations helped us understand the gaps and needs, which guided us in planning useful features for the system.

5.2.1.2 Admin

5.2.1.2.1 Login

5.2.1.2.1.1 Log into the system using username and password.

5.2.1.2.2 User Management

5.2.1.2.2.1 View all users in the system including assistant, buyers, sellers, warehouse staff, and delivery men with details such as name, email, phone number address, role and username.

5.2.1.2.2.2 Update the information of users such as address, name etc. whenever necessary.

5.2.1.2.2.3 Delete users from the system.

5.2.1.2.2.4 Approve or reject seller accounts after verifying their registration details like citizenship id, email verification including OTP, business registered certificate.

5.2.1.2.2.5 Add staff members who will help in managing inventory and order Processing.

5.2.1.2.2.6 Add delivery men responsible for delivering products to customers.

5.2.1.2.2.7 Add an assistant who will help admin in tasks like seller management, product management and order management.

5.2.1.2.3 Product Management

5.2.1.2.3.1 Add categories to manage the product properly according to category and sub categories.

5.2.1.2.3.2 Flag/ remove the inappropriate items uploaded by the seller.

5.2.1.2.4 Order & Earning Management

5.2.1.2.4.1 Monitor all orders placed by buyers across the platform including order

details, product information, payment method, and order status.

5.2.1.2.4.2 Assign delivery tasks to delivery men for delivering the products to buyers.

5.2.1.2.5 Promotional Settings

5.2.1.2.5. 1 Create and manage promotional campaigns such as seasonal sales, festival offers, or special discounts

5.2.1.2.6 Staff Management

5.2.1.2.6.1 Assign staff with certain roles such as assistant and warehouse staff.

5.2.1.2.7 Location

5.2.1.2.7.1 Admin can add and manage delivery locations where the platform services are available.

5.2.1.2.8 Logout

5.2.1.2.8.1 Admin can logout from the system.

5.2.1.3 Assistant

5.2.1.3.1 Login

5.2.1.3.1.1 Login with valid credentials provided by admin such as username & password .

5.2.1.3.2 User Management

5.2.1.3.2.1 View all users in the system including buyers, sellers, warehouse staff , and delivery men with details such as name, email, phone number address, role and username.

5.2.1.3.2.2 Update the information of users such as address, name etc. whenever necessary.

5.2.1.3.2.3 Approve or reject seller accounts after verifying their registration details like

citizenship id, email verification including OTP, business registered certificate.

5.2.1.3.3 Product management

5.2.1.3.3.1 Add categories to manage the product properly according to category and sub categories.

5.2.1.3.3.2 Flag/ remove the inappropriate items uploaded by the seller.

5.2.1.3.4 Order & Earning Management

5.2.1.3.4.1 Monitor all orders placed by buyers across the platform including order details, product information, payment method, and order status.

5.2.1.3.4.2 Assign delivery tasks to delivery man for delivering the products to buyers.

5.2.1.3.5 Emergency Fast Delivery

5.2.1.3.5.1 When a buyer selects emergency fast delivery, the assistant gets an immediate notification about the urgent order so they can process it quickly.

5.2.1.3.5.3 Assigns the nearest delivery men to the order and gives it priority.

5.2.1.3.5.4 After the order is delivered, the system marks the status as Delivered

Emergency Order.

5.2.1.3.6 Logout

5.2.1.3.6.1 Assistant can logout from the system.

5.2.1.4 Warehouse Staff

5.2.1.4.1 Login

5.2.1.3.4.1.1 Log in to the system using their assigned username and password.

5.2.1.4.2 Inventory Management

5.2.1.3.4.2.1 View the available stock of products including product name, quantity, and availability status.

5.2.1.4.2.2 Update stock quantities when new products arrive or when products are dispatched.

5.2.1.4.2.3 Mark product availability to ensure only stock items are shown to buyers.

5.2.1.4.3 Order Processing

5.2.1.4.3.1 View assigned orders that need to be packed/dispatched.

5.2.1.4.4 Logout

5.2.1.4.4.1 Staff can logout from the system.

5.2.1.5 Seller**5.2.1.5.1 Registration and Login**

5.2.1.5.1.1 Register with name, email, contact number, citizenship id, registered business certificate to use different functionalities.

5.2.1.5.1.2 Login into the system using valid credentials like username and password.

5.2.1.5.2 Product Management

5.2.1.5.2.1 Add new products by providing information such as product name, description, price, category, and product images.

5.2.1.5.2.2 Update the product and wait for approval from admin.

5.2.1.5.2.3 Set price, edit/update the price accordingly.

5.2.1.5.2.4 Upload images of the product.

5.2.1.5.3 Order Management

5.2.1.5.3.1 View orders placed by buyers for their products.

5.2.1.5.3.2 Accept/reject order placed by the buyer.

5.2.1.5.3.5 Mark order as ready for delivery after packaging.

5.2.1.5.4 Customer Engagement

5.2.1.5.4.1 Respond to buyer reviews, comments, and product inquiries.

5.2.1.5.5 Logout

5.2.1.5.5.1 Seller can logout from the system.

5.2.1.6 Buyer

5.2.1.6.1 Sign up and Login

5.2.1.6.1.1 Sign up with name, email, address to use different functionalities.

5.2.1.6.1.2 Login into the system.

5.2.1.6.2 Shopping Features

5.2.1.6.2.1 Browse the product based on categories, price and availability.

5.2.1.6.2.2 Search product based on categories.

5.2.1.6.2.3 Proceed to checkout using available payment methods including cash on delivery.

5.2.1.6.3 Order Management

5.2.1.6.3.1 Add products to the cart before placing an order.

5.2.1.6.3.2 View and track order status such as shipped, out for delivery, or delivered.

5.2.1.6.4.3 Allows buyers to select **Emergency Fast Delivery** for urgent orders so that the product can be delivered faster than normal delivery.

5.2.1.6.4 AI-Powered Experience

5.2.1.6.4.1 Personalized product recommendations based on buyer browsing, purchase history, context and budget based.

5.2.1.6.4.2 Displays recently viewed products, recommended for you and frequently purchased products.

5.2.1.6.5 AI Smart Comparison

5.2.1.6.5.1 Allows buyers to compare multiple products based on price and key features.

5.2.1.6.5.2 Use basic AI techniques to highlight the product that offers the best value among the compared items.

5.2.1.6.5.3 Helps buyers make informed purchasing decisions by presenting product differences clearly.

5.2.1.6.5 Provide Comment

5.2.1.6.5.1 Create an account to buy the product to provide feedback.

5.2.1.6.5.2 Reply to the comments on the product.

5.2.1.6.5.3 Detects positive and negative comments.

5.2.1.6.6 Logout

5.2.1.6.6.1 Buyer can logout from the system.

5.2.1.7 Delivery Men

5.2.1.7.1 Login

5.2.1.7.1.1 Log in using their assigned username and password.

5.2.1.7.2 Delivery Management

5.2.1.7.2.1 View assigned deliveries assigned by the admin.

5.2.1.7.2.2 Accept/reject delivery tasks.

5.2.1.7.2.3 View delivery details including order ID, pickup location, and delivery address.

5.2.1.7.3 Order Tracking

5.2.1.7.3.1 Mark orders as Picked Up after collecting products from the warehouse.

5.2.1.7.3.2 Update delivery status as Out for Delivery or Delivered

5.2.1.7.4 Earnings Management

5.2.1.7.4.1 View completed deliveries.

5.2.1.7.4.1 Track earnings based on completed delivery tasks.

5.2.1.7.5 Logout

5.2.1.7.5.1 Delivery men can logout from the system.

5.1.2.2 Non-Functional Requirements

- **Performance:** The performance of the website depends upon the system specification and network strength.
- **Security:** The details of the users are well managed and accountable to every user.
- **Scalability:** The system can be extended later due to well managed codes and documents.
- **Maintainability:** The system is easy to maintain.

5.3 Feasibility study

Feasibility study analyzes whether the project is likely to succeed. The study identifies possible issues and problems that could arise while development and operation. Feasibility study is designed to help decision-makers determine whether or not a proposed project or investment is likely to be successful. It identifies both the known costs and the expected benefits.

5.3.1 Technical Feasibility

Our system is designed as a web-based platform, making it accessible from any browser and independent of specific devices or operating systems. Moreover, it is developed using standard software engineering principles and an object-oriented programming approach, allowing easy updates, addition of new features, and modules as per requirement.

Technical feasibility checks whether the proposed project can be implemented using the available technology and resources. Here are some points to consider for technical feasibility:

- Availability of required technology
- Compatibility with web browsers and devices
- Scalability to handle more users and products in the future
- Security to protect user and payment data.

The technical feasibility assessment confirms that the proposed project is feasible. It can use existing technology effectively, adapt to future upgrades and ensure secure, reliable operation, making it a viable solution for implementation.

5.3.2 Operational Feasibility

Operational feasibility focuses on determining whether the proposed system can work effectively within the organization's operations and processes. It evaluates whether the system will be accepted by users and whether it can be operated smoothly in a real environment. Some important considerations for operational feasibility include:

- Resource Availability
- Cost-effectiveness
- Organizational Impact

5.3.3 Economic Feasibility

Economic feasibility is the process of determining whether a project is financially worthwhile. A feasibility report includes a cost/benefit analysis of the project. It also forecasts an expected return on investment(ROI), as well as outlines any financial risks. The goal at the end of the economic feasibility study is to understand the economic benefits the project will drive.

- **Cost estimation:**

Software cost is only a small part of the total cost of the whole computer system. Many things can affect the final cost, such as the people working on the project, the technology used, and the hardware and software needed.

Table 1: Total cost estimation

S.N.	Category	Sub-category	Estimated Cost (in Rs)
1	Infrastructure Cost	a) Software Cost	4000
		b) Hardware Cost	5000
2	Effort Cost		12000
3	Overhead/AMC	a) Training	5000
		b) AMC	4000
Total Cost			30000

Overall, the cost estimation starts from infrastructure costs (software and hardware), followed by the development effort cost, and finally the training and annual maintenance costs.

Altogether, the total estimated cost of the system is Rs. 30,000, which supports the successful development and maintenance of the system.

- **Payback period**

The payback period is the time required to recover the initial cost of an investment. It is the number of years it would take to get back the initial investment made for a project. The payback period will be used to compare projects and derive the number of years it takes to get back the initial investment.

Formula:

$$\begin{aligned}\text{Payback period} &= \frac{\text{Cost of Investment}}{\text{Average annual cash flow}} \\ &= 30000/48450 \\ &= \approx 7.5 \text{ months}\end{aligned}$$

In our system, the estimated payback period of the system is about ≈ 7.5 months. This indicates that the initial investment required for developing and implementing the system can be recovered within a certain period of time. Therefore, the project is considered financially feasible and low-risk, as the investment can be returned quickly through the expected revenue generated by the system.

- **Return on investment (ROI)**

Return on Investment (ROI) is a financial metric used to measure the profitability and efficiency of an investment. It quantifies the return or gain generated from an investment relative to its cost. Return on Investment (ROI) indicates that the project is highly profitable, generating a return to the initial investment. ROI reflects efficient use of investment funds and suggests that the project is a good financial choice.

The ROI formula is as follows:

$$\text{ROI}(\%) = (\text{Gain from investment} - \text{Cost of investment}) / \text{Cost of investment} * 100$$

$$\text{ROI}(\%) = (48450 - 30000) / 30000 * 100$$

$$\text{ROI}(\%) = 18450 / 30000 * 100$$

$$\text{ROI}(\%) \approx 61.5\%$$

From the above calculation, the Return on Investment (ROI) is $\approx 61.5\%$, which indicates that the project can generate a good profit compared to the initial investment. This shows that the investment is financially beneficial and the project is a profitable and efficient use of resources.

• **Annual Maintenance Cost(AMC)**

Annual Maintenance Cost is the fee provided by the buyers to the manufacturers for maintenance of the system and for web hosting. The prices may vary according to the added additional features of the system. The estimated annual maintenance cost is Rs. 5000.

5.3.4 Schedule Feasibility

Schedule feasibility is defined as the probability of a project to be completed within its schedule time limits, by a planned due date. If a project has a high probability to be completed on-time, then its schedule feasibility is appraised as high. This assessment is the most important for project success. In scheduling feasibility, an organization estimates how much time the project will take to complete.

Gantt Chart

PROCESS	3-Mar	31-Mar	22-Apr	12-May	11-Jul	2-Aug
Requirement Analysis	28 days					
System Analysis		22 days				
System Design			20days			
Coding				60 days		
Testing & Debugging					22 days	
Deployment						13 days
Maintenance						15 days
Documentation	180 days					

Figure 1: Gantt Chart

Table 2: Details of Schedule

Activities	Start Date	End Date	Duration
Requirement Analysis	3-Mar	30-Mar	28
System Analysis	31-Mar	21-Apr	22
System Design	22-Apr	11-May	20
Coding	12-May	10-Jul	60
Testing & Debugging	11-Jul	1-Aug	22
Deployment	2-Aug	14-Aug	13
Maintenance	15-Aug	29-Aug	15
Documentation	3-Mar	29-Aug	180

5.4 Tools Used

5.4.1 Front End Tools

HTML5:

HTML5 is the latest version of Hypertext Markup Language used to structure and display content on the web. In SajiloMart, HTML5 is used to create clear, organized, and responsive web pages, ensuring a smooth experience for users on different devices.

CSS3:

CSS3 (Cascading Style Sheets) is used to style and design web pages. In SajiloMart, CSS3 helps make the interface visually appealing, consistent, and modern, while reducing development time.

JavaScript:

JavaScript is a lightweight, versatile scripting language mainly used to make web pages interactive and dynamic. In SajiloMart, it is used to add features like product search, order updates.

5.4.2 Back End Tools

• SQLite

SQLite is a lightweight, open-source database that stores all data in a single file. It is easy to set up, reliable, and well-suited for small to medium-scale projects. In the SajiloMart project, SQLite is used to store and organize user information, product catalogs, order histories, vendor records, delivery tracking, and payment transactions efficiently.

• Django

Django is the main backend framework for the SajiloMart platform. It follows the Model-Template-View (MTV) architecture, which separates business logic, data, and user interface for better organization. Django handles routing, form processing, user

authentication, and database interactions using its built-in tools, which makes development faster and more structured. It connects seamlessly with SQLite to manage users, products, orders, reviews, and payments. Django's focus on security, scalability, and clean design provides a strong foundation for building and maintaining the SajiloMart e-commerce platform using Python.

6. Structuring System Requirements

Structuring system requirements focuses on defining, organizing, and understanding data and its relationships. This process is crucial for designing databases, programs, user interfaces, and reports, ensuring data accuracy and consistency in an information system.

6.1 System Flowchart Diagram

The flowcharts are simple visual tools that help us understand and represent processes very easily. They use shapes like arrows, rectangles, and diamonds to show steps and decisions clearly. If someone is making a project or explaining a complex task, flowcharts can make complex ideas easier to understand.

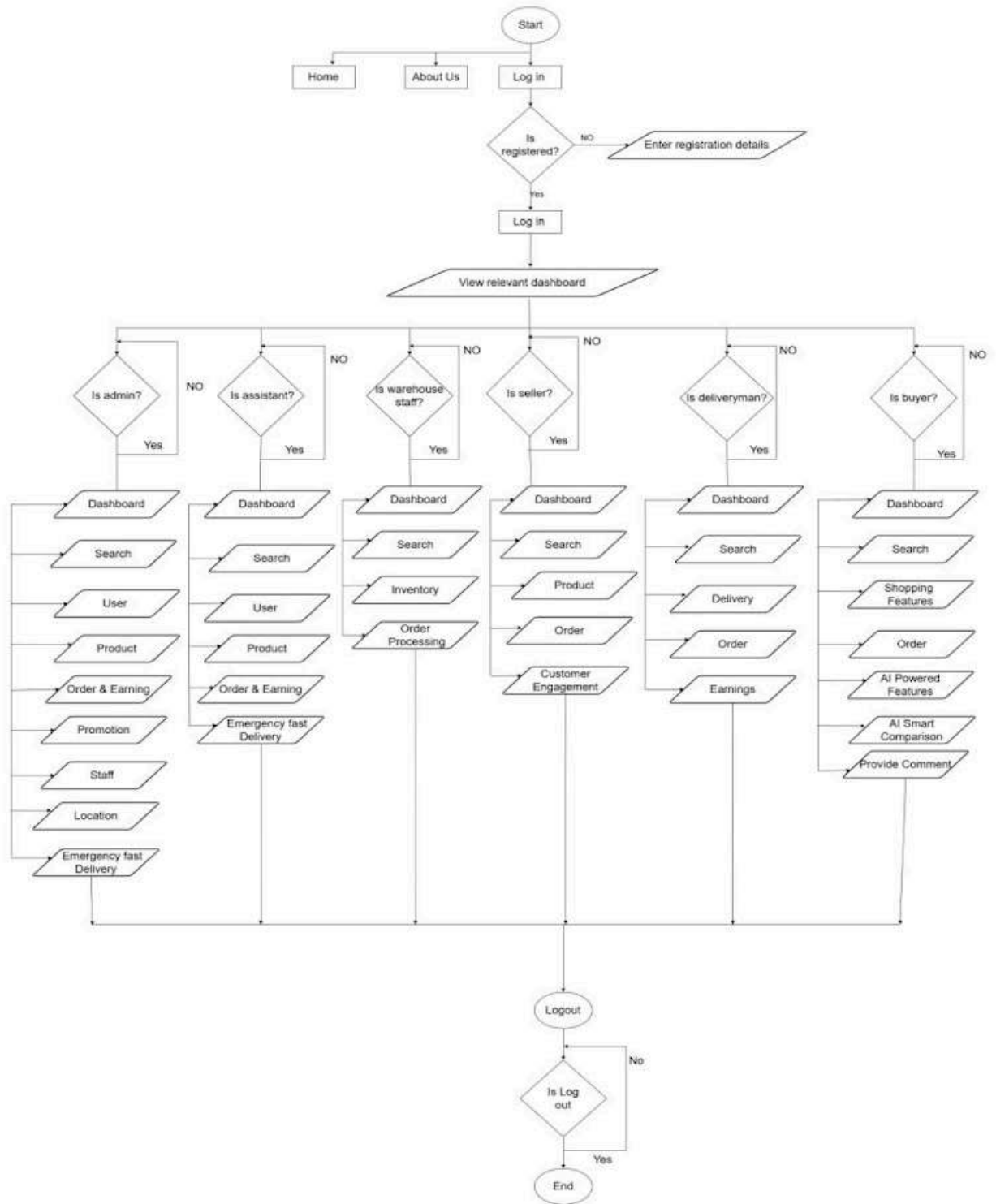


Figure 2: System Flow chart

6.2 Use case Diagram

A Use Case Diagram is a type of UML diagram that illustrates how users, known as actors, interact with a system to achieve specific goals. It focuses on the system's functional requirements, showing the different use cases or tasks the system must perform and the relationships between actors and these use cases. Actors can be people, external systems, or devices that communicate with the system.

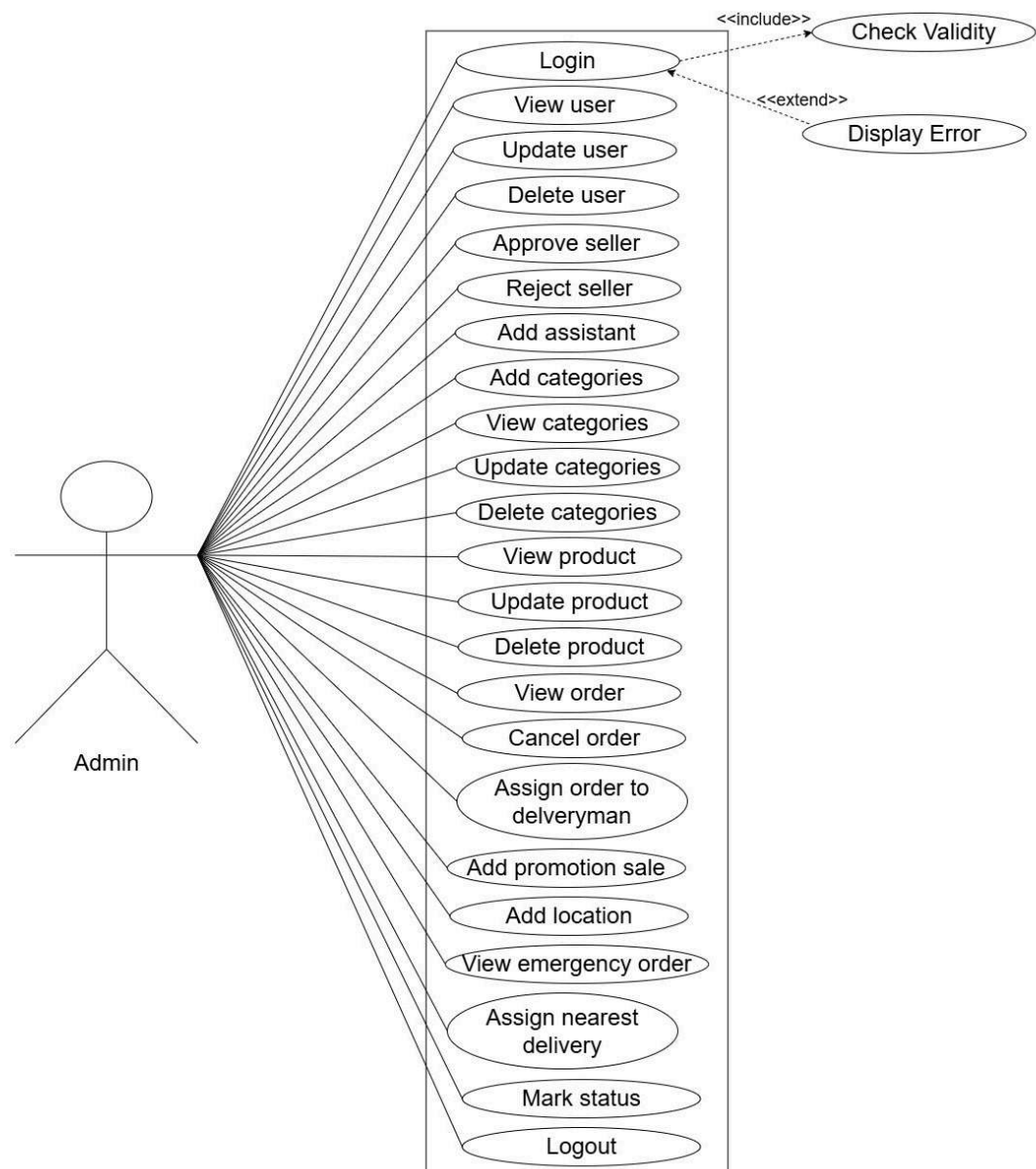


Figure 3: Use case diagram of Admin

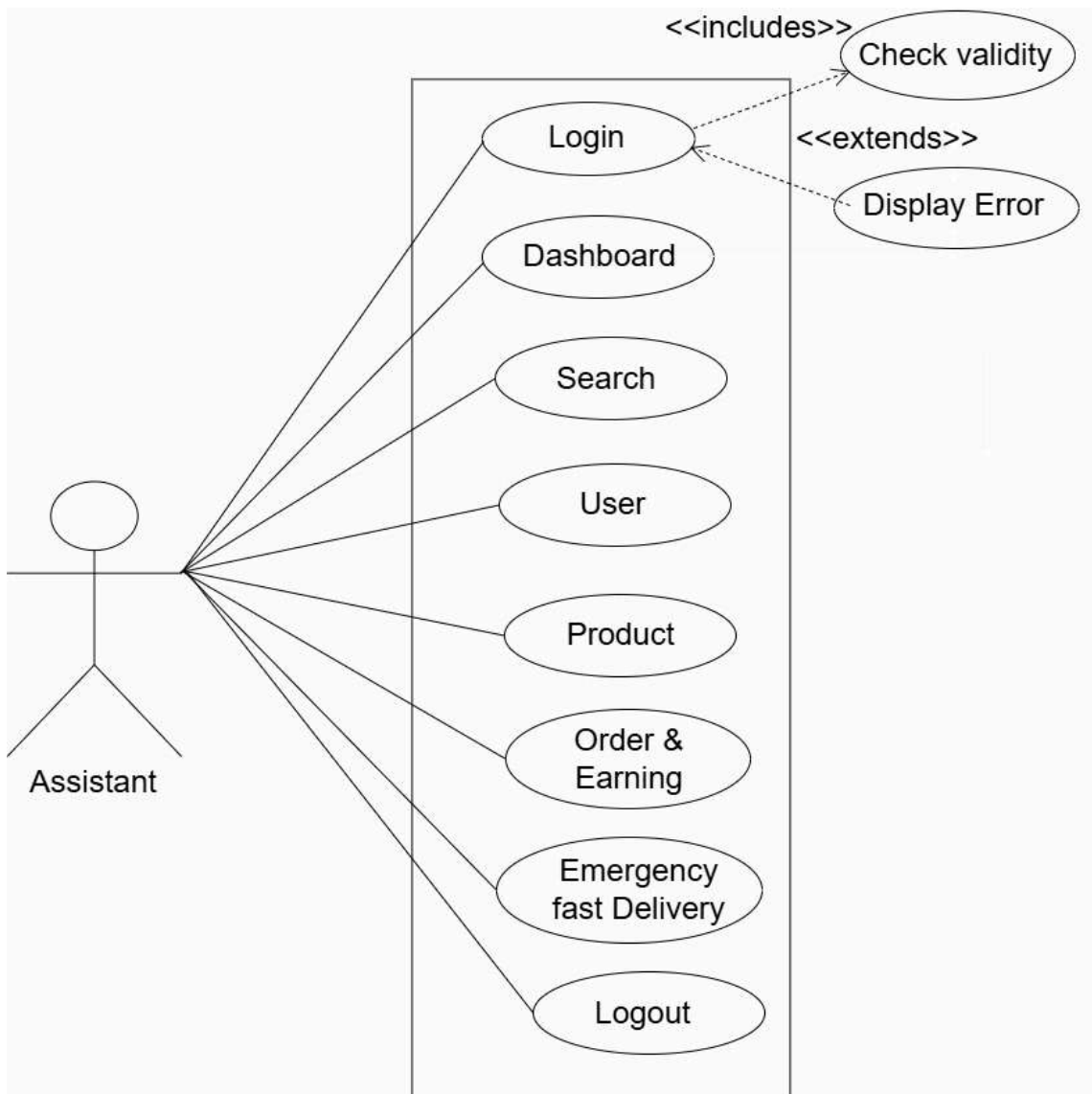


Figure 4: Use case diagram of Assistant

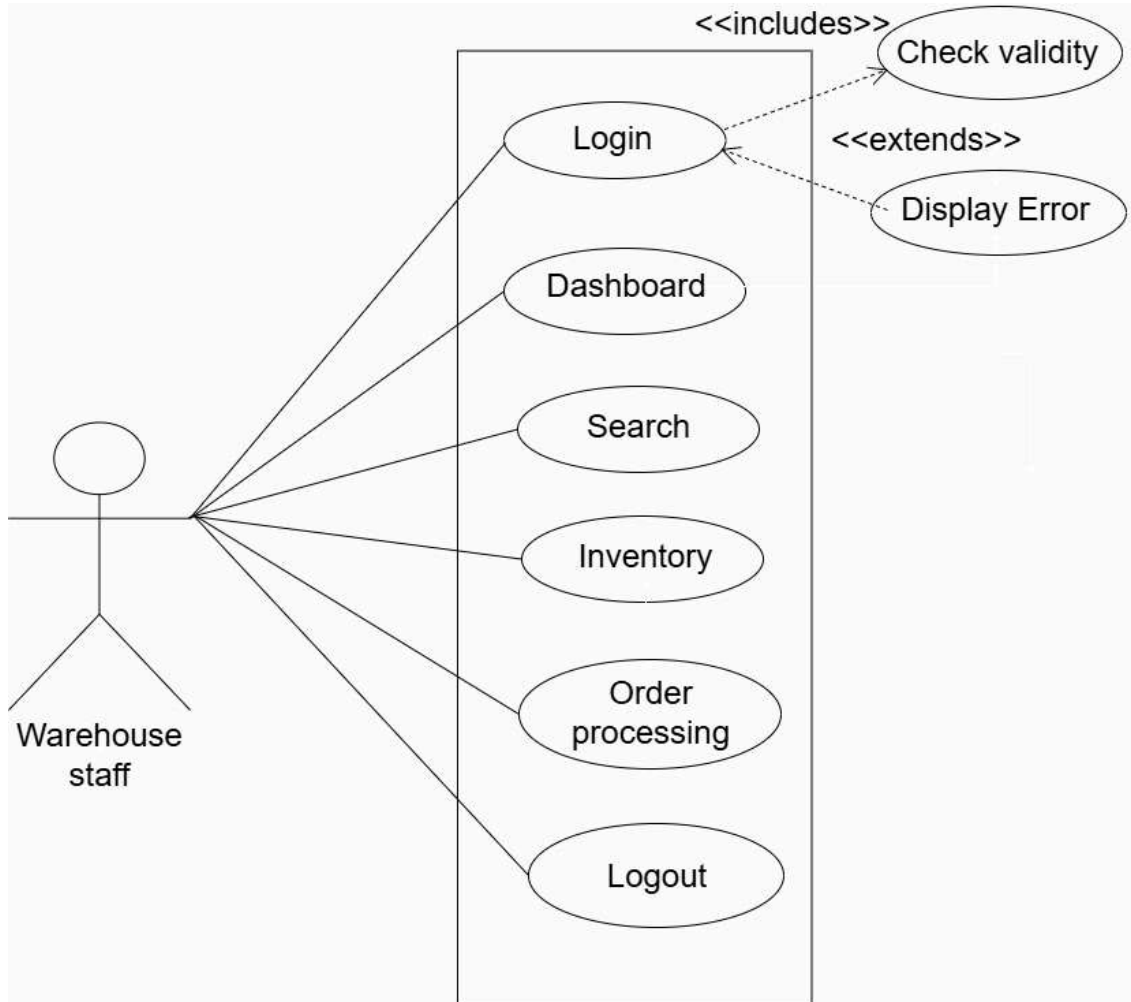


Figure 5: Use case diagram of Warehouse Staff

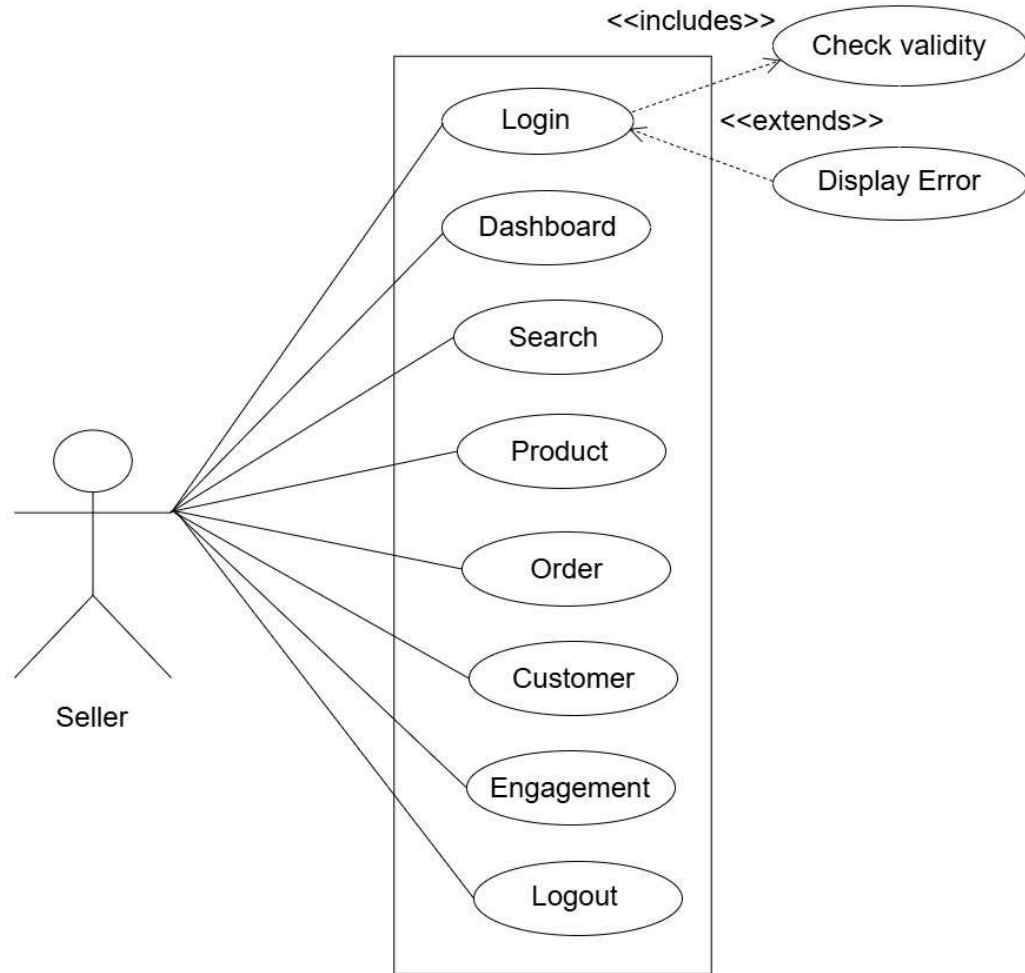


Figure 6: Use case diagram of Seller

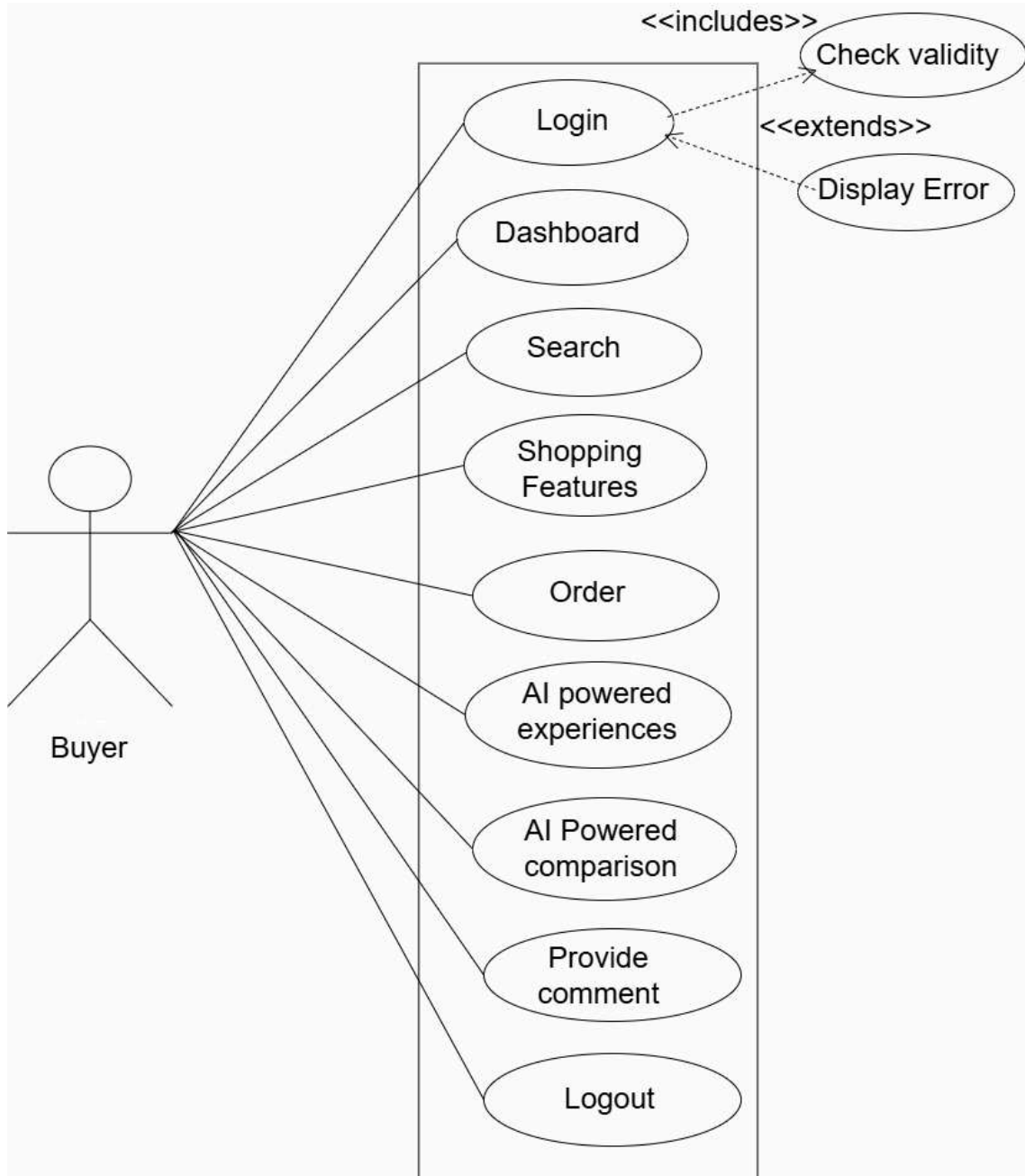


Figure 7: Use case diagram of Buyer

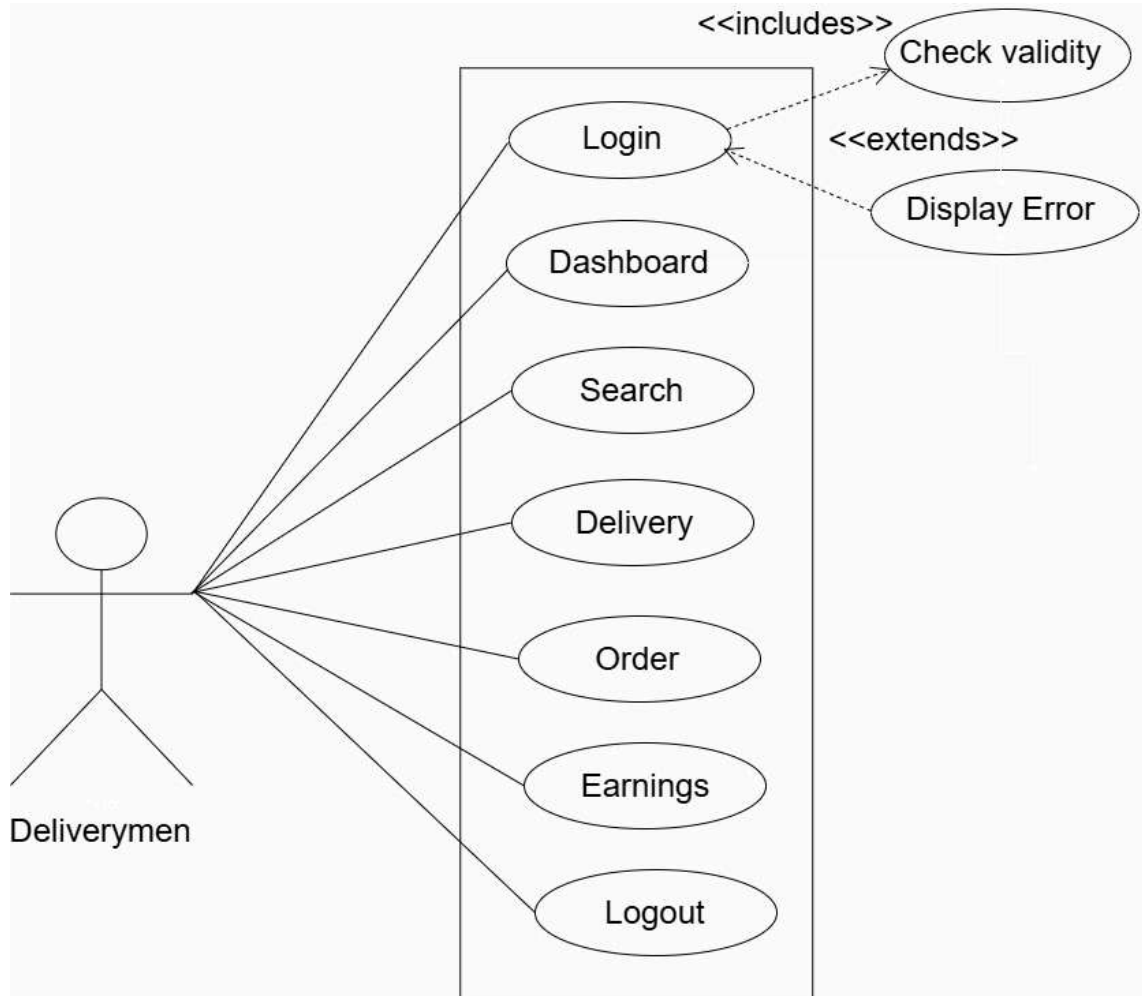


Figure 8: Use case diagram of Delivery Men

6.3 Entity Relationship Diagram

An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research. They use a defined set of symbols such as rectangles, diamonds, ovals and connecting lines to depict the interconnectedness of entities, relationships and their attributes.

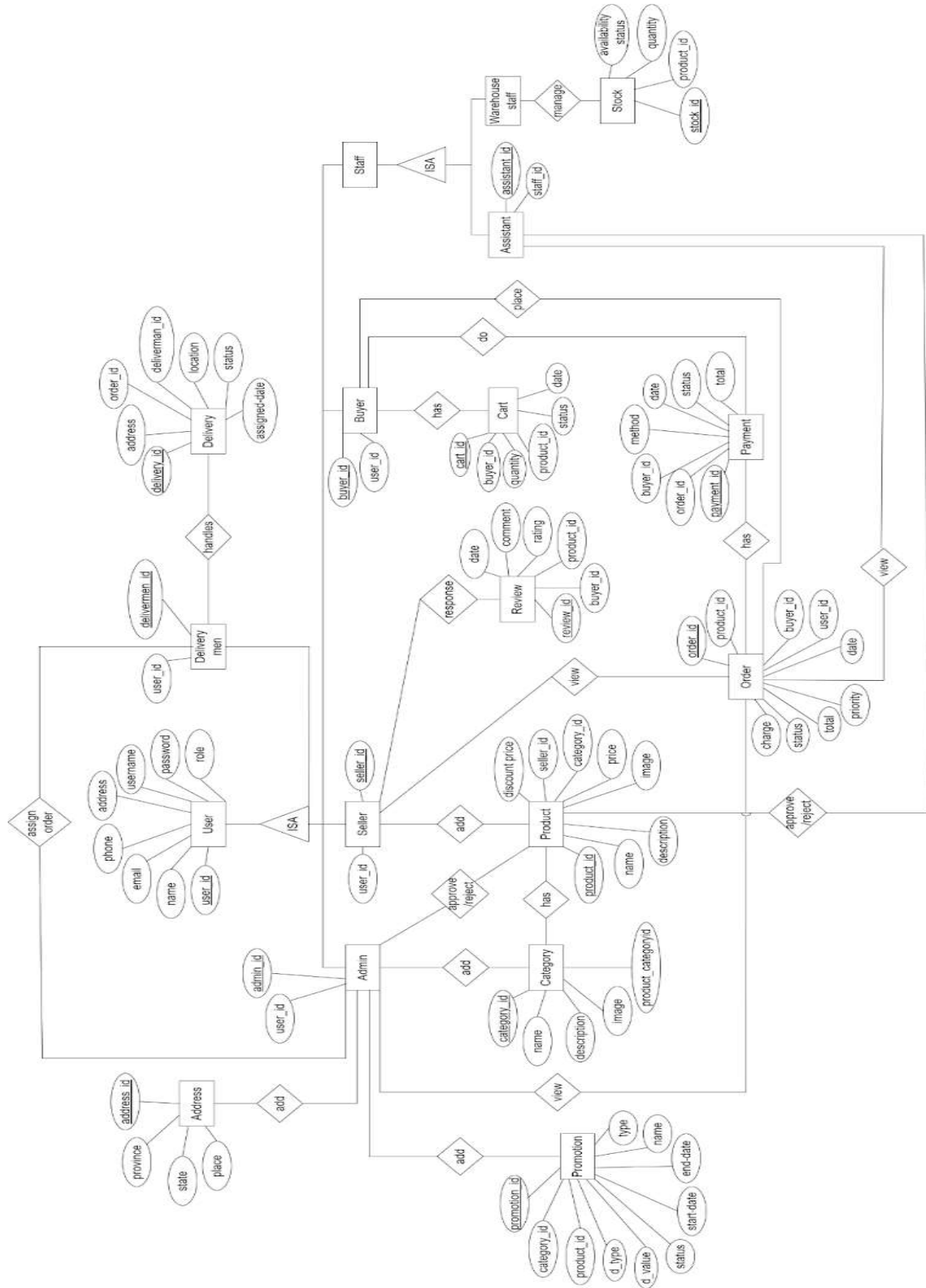


Figure 9: Entity Relationship Diagram

6.4 DFD Diagram

Data Flow Diagram is a visual representation of the flow of data within the system. It helps to understand the flow of data throughout the system, from input to output, and how it gets transformed along the way. The models enable Admin, Assistant, Warehouse Staff, Seller, Buyer and Delivery men to work together effectively during the analysis and specification of requirements.

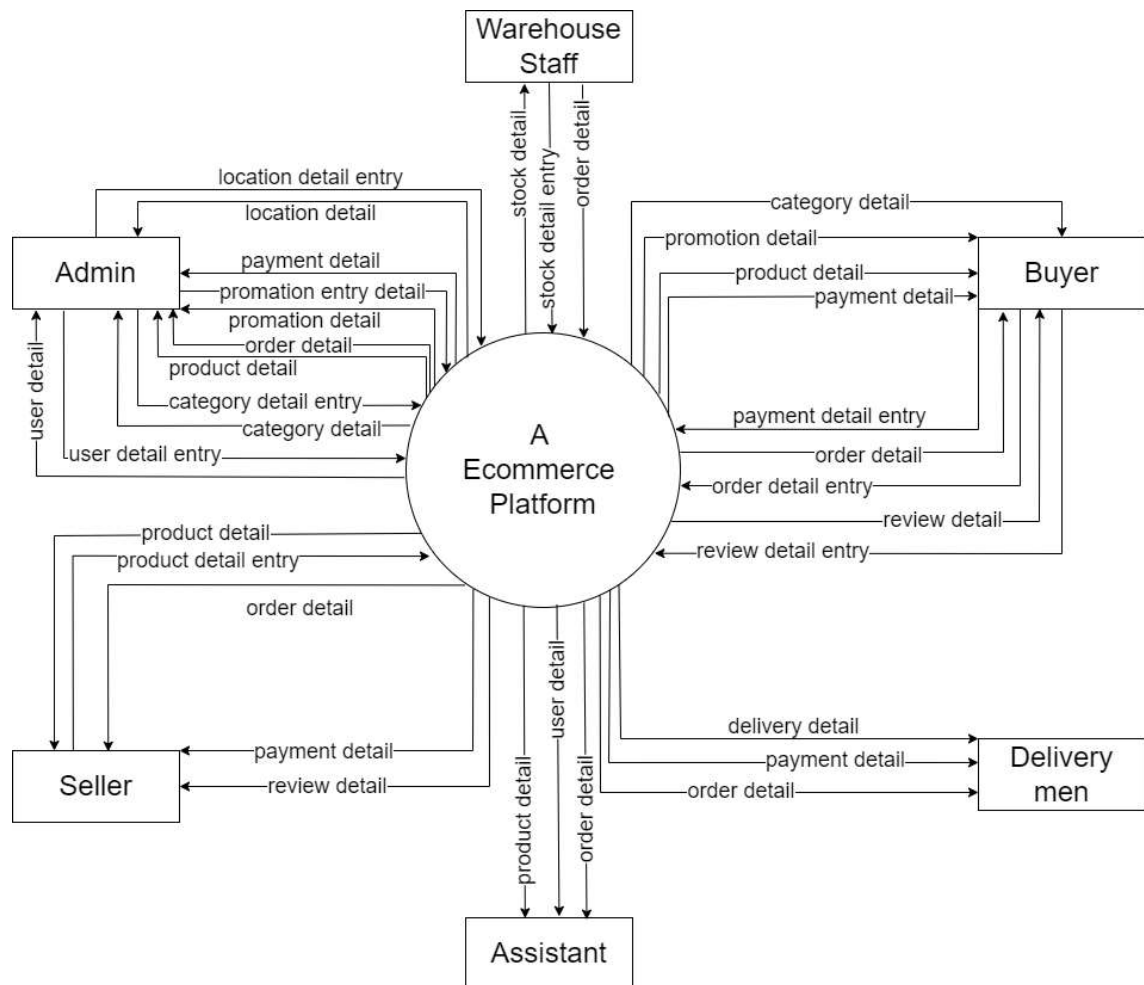


Figure 10: Data Flow Diagram (level 0)

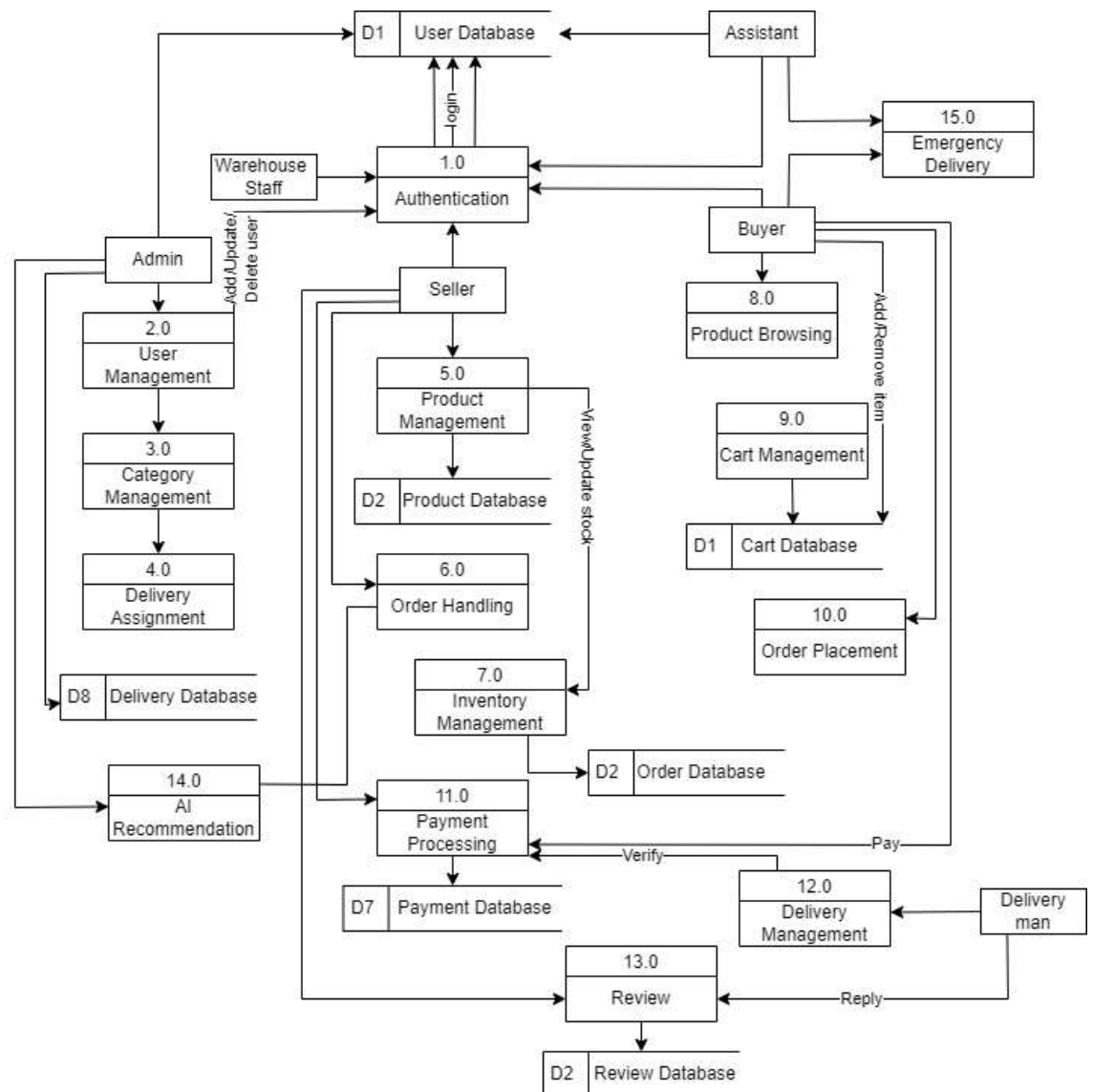


Figure 11: Data Flow Diagram (level 1)

6.6 Class Diagram

A class diagram is a type of UML (Unified Modeling Language) diagram that shows the structure of a system by displaying its classes, attributes, methods, and relationships. It helps visualize how objects interact and how data is organized in a program. Class diagrams are widely used in object-oriented design to plan and document

software before coding. They make it easier to understand system architecture and relationships between different parts.

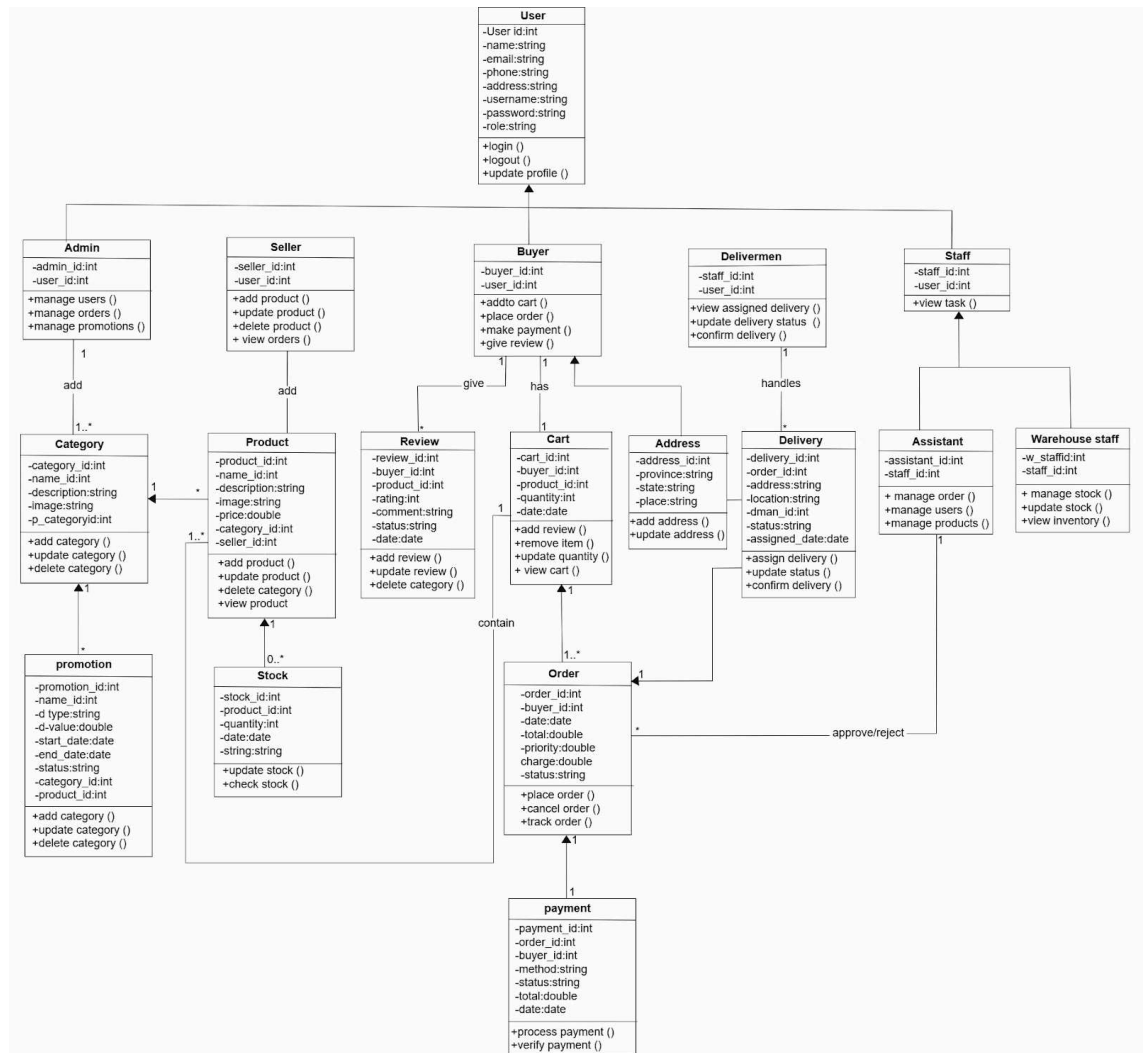


Figure 12: Class Diagram

7. Expected Outcome

The system SajiloMart is expected to make online shopping easier and more convenient for customers. It will allow users to browse products, place orders, and make payments in a simple and organized way. Sellers will be able to list their products, manage

orders, and reach more customers through the platform. Buyers can compare products, check details, and choose items according to their needs and budget. Admin will manage the system and ensure that everything runs smoothly. Overall, the system will save time, improve the buying and selling process, and provide a reliable online shopping experience for users.